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Figure 1: Mount Rainier, Washington. Photograph by Yoni Rodriguez, 2022.

## SHM 102: Occupational Health

### Introduction to Occupational Health

#### Lecture 01 · Fall 2026

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## What is Occupational Health?

### Before We Define It...

Let's start with what you already think.

### Poll Everywhere

1. *What comes to mind when you hear the term “occupational health”?*
2. *In your own words, how would you define occupational health?*

There is no expectation that you know the formal definition yet. Give me your best answer.



### A working definition

*“Occupational and environmental health is the study and prevention of health problems caused or influenced by the environments where people work, live, and interact.”* - **Centers for Disease Control and Prevention (CDC)**

### **Where does Occupational Health occur?**

- Do you get exposed on your way to work?
- Does the community nearby get exposed?
- Do you take the exposures home?

### **Why is Occupational Health important?**

- Does this impact my health?
  - Does this impact the health of my family?
  - Does this impact the health of my community?
-

## ***This Pesticide May Be Too Dangerous to Use. Farmers Say They Need It.***

California announced that it's phasing out paraquat, a pesticide widely used in the United States that is linked to Parkinson's disease.



American farmers spray millions of pounds of paraquat on crops each year, though experts could not say precisely how much. Bill Barksdale/Design Pics Editorial/Universal Images Group, via Getty Images

*The New York Times*, August 13, 2026.

[Read the article →](#)

## Where is paraquat used?

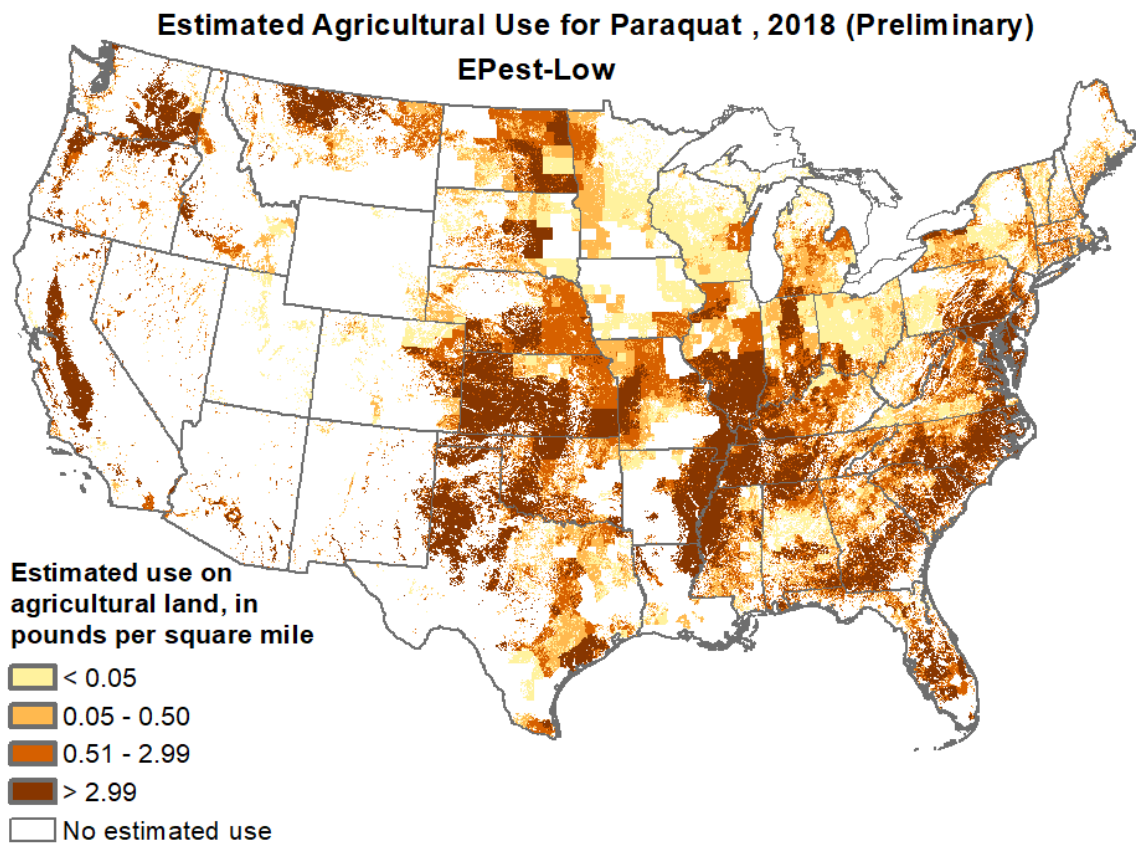


Figure 2: Estimated agricultural use of paraquat in the United States, 2018.

Source: U.S. Geological Survey.

[Explore paraquat use map →](#)

## Who is exposed?



Figure 3: Pesticide application with fluorescent tracer.

Where was the applicator exposed?



Figure 4: Fluorescent tracer visible on the applicator's face following pesticide application.



Figure 5: Fluorescent tracer visible on the applicator's hand following pesticide application.

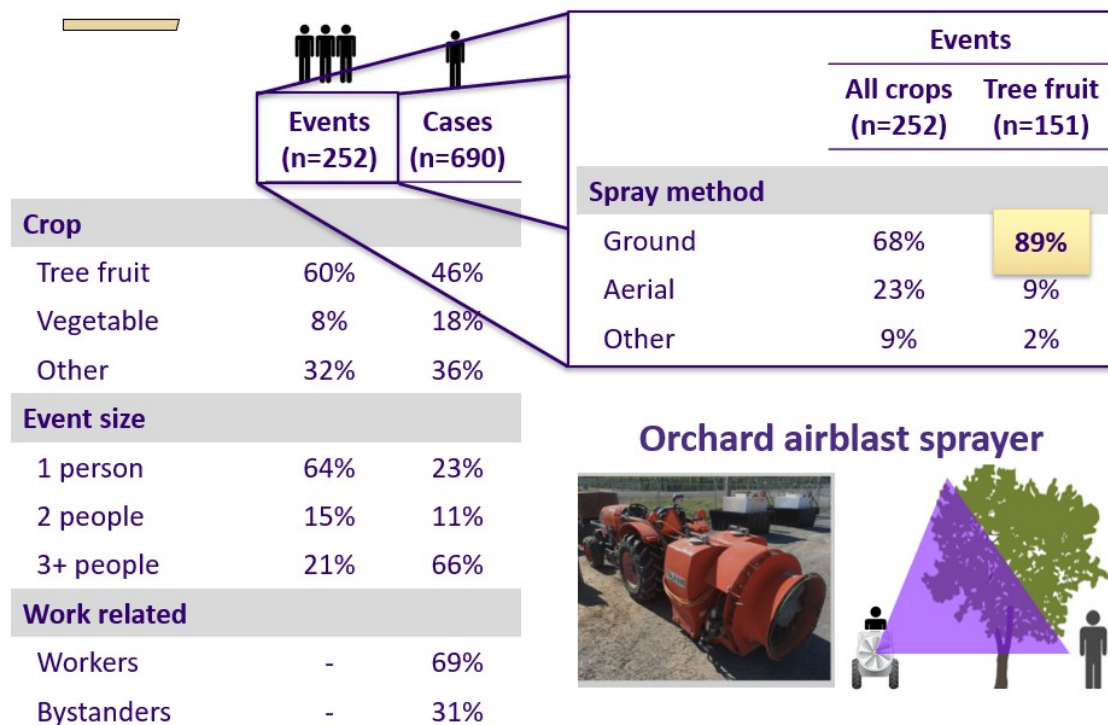
**Photos:** Cesar Ramirez and Matteo Vijarro.

**Research supported by:** [Pacific Northwest Agricultural Safety and Health Center \(PNASH\)](#), University of Washington.

## Is the applicator the only person exposed?

### Pesticide drift in Washington State

#### Pesticide drift illness statistics, Washington, 2000-2015



Source: Washington Department of Health (WA DOH)

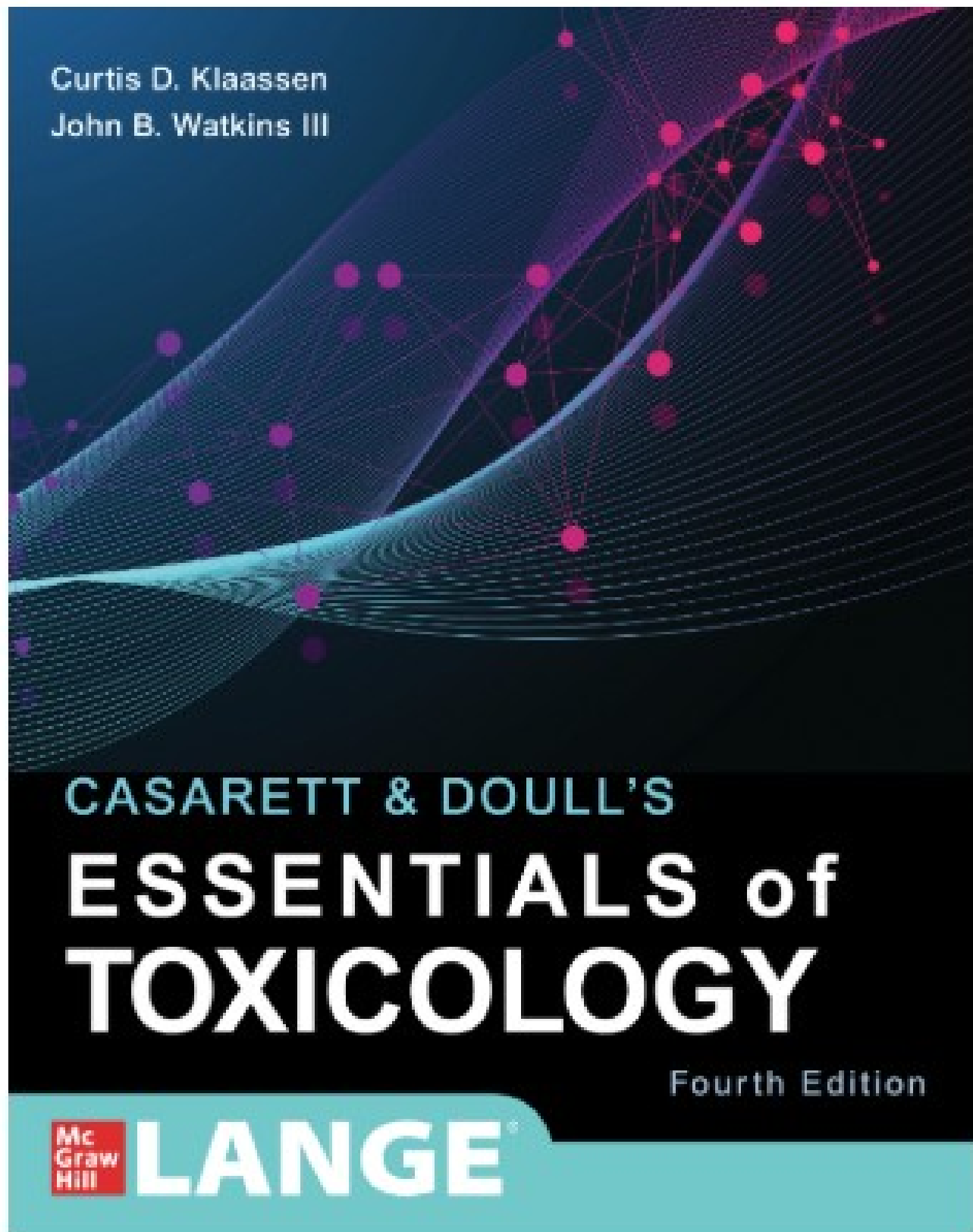
Source: Kasner et al. (2021), *Environmental Health*.

[View study on PubMed →](#)



What happens when exposure occurs?

Paraquat: acute toxicity



## **Bipyridyl Compounds**

**Paraquat** is a fast-acting, nonselective contact herbicide used to control broad-leaved weeds and grasses in plantations and fruit orchards, and for general weed control.

**The herbicide has a high acute toxicity.** Paraquat is poorly metabolized and excreted in the urine. **Paraquat accumulates in lung tissue**, particularly, in Type II alveolar epithelial cells and Clara cells. It has minimal to no genotoxicity, is not carcinogenic in rodents, has no effect on fertility, is not teratogenic, and only produces fetotoxicity at maternally toxic doses. Paraquat can be reduced to form a free radical with concomitant production of superoxide anion (Fig. 22-6). Intracellular redox cycling results in the oxidation of NADPH, leading to its cellular depletion, which is augmented by the detoxification of hydrogen peroxide formed in the glutathione peroxidase/reductase enzyme system to regenerate GSH.

*Source: Costa (2021), "Toxic Effects of Pesticides," in Casarett & Doull's Essentials of Toxicology.*  
[View textbook chapter →](#)

### **What can severe acute exposure do?**

**Upon acute exposure to lethal doses of paraquat, damage to alveolar epithelial cells is seen within 24 hours after exposure** and progresses in the following 2 to 4 days with large areas of the alveolar epithelium completely lost. This is followed by alveolar edema, extensive infiltration of inflammatory cells into the alveolar interstitium, and finally death due to severe anoxia. Survivors of this destructive phase show a subsequent

*Source: Costa (2021), "Toxic Effects of Pesticides," in Casarett & Doull's Essentials of Toxicology.*  
[View textbook chapter →](#)

What about long-term health?

## Environmental Factors Relevant to Neurodegenerative Diseases

The relationship between MPTP intoxication and parkinsonism stimulated investigations into the role that environmental and occupational exposures may play in the pathogenesis of neurodegenerative diseases. Epidemiological studies implicate environmental and occupational exposure to pesticides, metals, and solvents as risk factors for PD. Studies of agricultural workers and communities have associated higher PD incidence with exposure to specific pesticides such as organochlorines, paraquat, rotenone, and OP compounds. Converging evidence in PD research suggests that mitochondrial damage and oxidative stress are the primary causes of DAergic neuronal death.

Source: Moser et al. (2021), "Toxic Responses of the Nervous System," in Casarett & Doull's Essentials of Toxicology.

[View textbook chapter →](#)

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## What did we just do?

**We identified a hazard**

Paraquat

↓

**We asked where it is used**

Agricultural workplaces and environments

↓

**We identified who can be exposed**

Applicators → Coworkers → Bystanders → Communities

↓

**We examined how people are exposed**

Dermal contact → Inhalation → Ingestion

↓

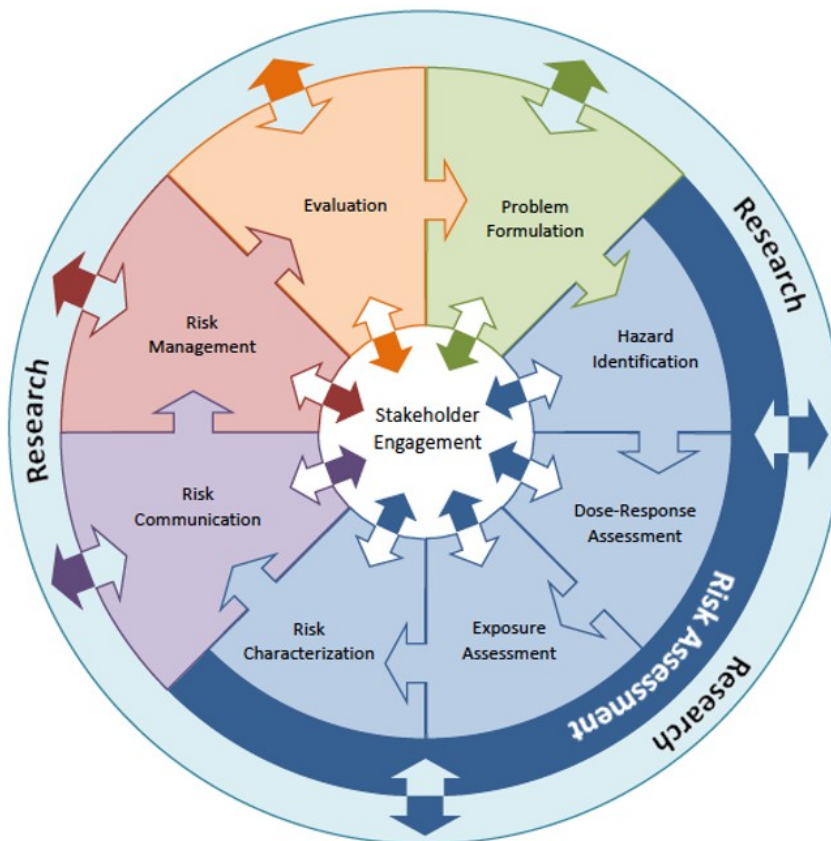
We considered what happens after exposure

Toxicity → Health effects → Health risks

This is Occupational Health

How to Apply Occupational Health

Environmental & Occupational Health Framework

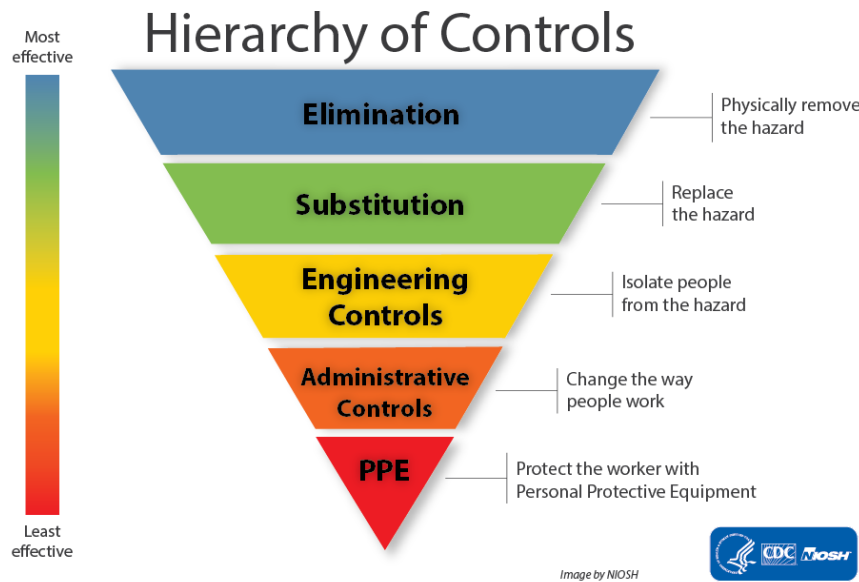


Source: University of Washington Department of Environmental & Occupational Health Sciences.  
[Explore the ENV H 501 course framework →](#)

## How do we control exposure?

If a hazard can affect health, What can we do about it?

## Hierarchy of Controls



Source: National Institute for Occupational Safety and Health (NIOSH).

[Explore the Hierarchy of Controls →](#)

## Apply it to our case

How could we reduce pesticide exposure?

Where would you intervene?

## Apply the Hierarchy of Controls

Take a couple of minutes to discuss with your classmates.

Using the **Hierarchy of Controls**, consider:

1. What could we do to reduce or prevent pesticide exposure?
2. Where in the hierarchy would your intervention fall?
3. Why would your approach reduce exposure?

When you're ready, submit your responses through Poll Everywhere.



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## What could prevention look like?

Can we measure environmental conditions and use that information before exposure occurs?

## Measuring the environment



Figure 6: Washington State University AgWeatherNet weather station.

*Source: Washington State University AgWeatherNet.*

[Explore AgWeatherNet →](#)

## From measurement to prediction

### Aim 1c and Aim 2 Results: Prosser N.E. ARIMA (2,1,1) Wind Speed Forecast. May 1<sup>st</sup> – June 1<sup>st</sup>, 2020

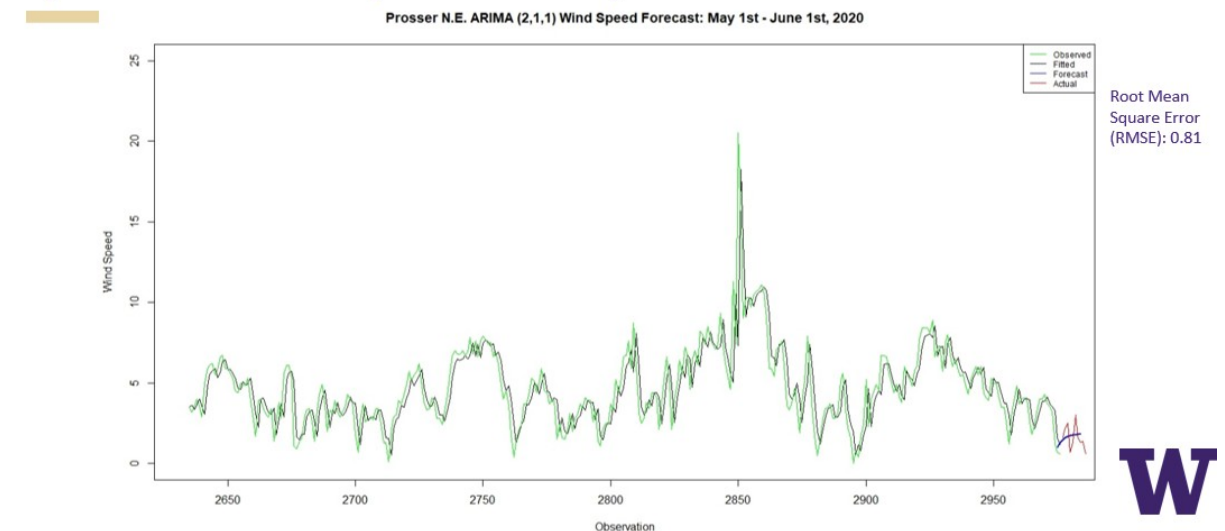


Figure 7: Short-term wind speed forecast using AgWeatherNet observations.

Source: Rodriguez, Y. — *Exploring Wind Ramping as a Determinant of Pesticide Drift*, University of Washington DEOHS.

[View research project →](#)

## Where does this fit in the Hierarchy of Controls?

**Administrative Controls**

Change the way people work

### Administrative Control

Forecast → Alert → Delay or reschedule application

**Engineering Controls**

Isolate people from the hazard

## Engineering Control

Forecast → Equipment responds → Modify or stop application

*Source: National Institute for Occupational Safety and Health (NIOSH).*

[Explore the Hierarchy of Controls →](#)

## What will we explore?

### Five broad categories of occupational hazards

Occupational Health

Chemical

Physical

Biological

Ergonomic

Psychosocial

## What will we learn in this course

| What we do              | Why it matters                                                                                                           |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>Identify Hazards</b> | Recognize conditions or agents that could cause harm so future or recurring exposures can be prevented.                  |
| <b>Measure Hazards</b>  | Collect accurate information about the hazard and its source so we can determine where and when exposure is occurring.   |
| <b>Assess Hazards</b>   | Understand the hazard, its source, the exposure context, and its significance within a specific workplace or setting.    |
| <b>Control Hazards</b>  | Develop effective interventions that eliminate or reduce the hazard or exposure using the <b>Hierarchy of Controls</b> . |

**Understand exposure. Prevent harm.**

## Key Terms

**These are the terms I expect you to know from today's lecture for the mid-term.**

